

Rajeev Mutyalu

Founder, Learn with Charlie • AI Workflows Architect • Creative Technology Lead • VFX & GenAI Systems Architect

☆ Founder, Learn with Charlie • GenAI Production Systems • 1917 • RRR • Mufasa

● Former Technicolor R&D Supervisor • Technicolor Fellowship Network Associate Member

UK Skilled Worker Visa • Existing Sponsorship in Place • Available Immediately

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🎯 Executive Summary

Founder of Learn with Charlie and former Technicolor R&D Supervisor with 20+ years of experience across visual effects production technology, software architecture, creative engineering and global engineering leadership. Founder of **Learn with Charlie**, a curriculum-bound AI maths tutor for Indian school children that teaches from their own textbook with natural voice, live blackboard, and zero guessing. Currently architecting Generative AI production systems at Astra Studios, including the **Studio.AI platform** integrating **Google Veo, Kling AI, Higgsfield, Seedance and ComfyUI node workflows** with character identity consistency, artist-facing controls and ACES colour conforming. Combines emerging generative video technology with established filmmaking workflows, DCC APIs including Nuke, Maya and Hiero, and production-grade OpenUSD systems.

Track record of taking emerging technologies from R&D into production, including the global **ReviewTool** platform caching 4K+ media in real-time, **ProNuke** modular shot assembly, **Meshroom photogrammetry** integration, and **Editorial Turnover, Conform & I/O systems**. Experienced in leading distributed engineering teams across London, Montreal and Bengaluru and working across engineering, artists, production and technical leadership across major productions including *1917* (Academy Award for Best Visual Effects), *RRR*, *Mufasa: The Lion King*, *Back in Action*, *Spaceman*, and *Prehistoric Planet*. Associate Member of the Technicolor Fellowship Network Committee and recipient of 9+ studio leadership & excellence awards.

Architecture & Technical Competencies

Generative Video

🎬 Generative AI Platforms & Workflows

- **Generative AI Platforms:** Google Veo, Kling AI, Higgsfield, Seedance and ComfyUI node workflows, latent consistency models.
- **Productionisation:** Studio.AI platform, controlled generation workflows supporting character and visual continuity, artist-facing controls and ACES colour conforming.

EdTech & AI Architecture

🚀 EdTech & AI Production Architecture (Learn with Charlie)

- **Learn with Charlie Architecture:** Founder of Learn with Charlie (learnwithcharlie.ai), curriculum-bound AI maths tutor for Indian school children; React, FastAPI, SymPy symbolic verification (zero guessing), and ElevenLabs voice.
- **Studio.AI & Production Systems:** Multi-model workflow integration, artist-facing controls, structured prompt routing, and connecting generative video outputs with established VFX pipelines.

Leadership

👥 Engineering Leadership & Delivery

- **Team & Technical Leadership:** Leading R&D software engineering, development, and TD teams across London, Montreal, and Bengaluru.
- **Production Execution:** Translating creative requirements into technical systems, taking R&D technologies into production, Agile/Scrum delivery.

Production Systems

🎥 VFX Production Systems & DCC APIs

- **Review & Assembly:** ReviewTool (real-time 4K+ media caching & multi-facility synchronisation), ProNuke modular shot assembly, Meshroom photogrammetry.
- **DCC & Studio APIs:** Nuke C++/Python API, Maya Python API, Hiero / Nuke Studio Python API, ShotGrid Python API, 3D Equalizer, Silhouette.

Agentic & MCP

🕴 Agentic AI & MCP Systems

- **Model Context Protocol:** Custom MCP server engineering, JSON-RPC 2.0, tool/resource schemas and pipeline API binding.
- **Agentic Workflows:** Claude Code, OpenAI Codex, Google Antigravity, subagent orchestration, task decomposition and multi-step tool workflows.

Private AI

🔒 Private / On-Premise AI

- **Local Inference:** Ollama, air-gapped studio inference, local private LLM deployments, GGUF / AWQ quantisation, context-window optimisation.
- **Secure Endpoints:** Secure internal AI endpoints designed to keep sensitive studio data within controlled infrastructure and reduce exposure to public AI services.

OpenUSD

📦 OpenUSD Production Architecture

- **USD Architecture:** OpenUSD (USD Python API, composition arcs, layer stacking, edit targets, sublayering), full-CG & live-action shot pipelines.
- **Colour & Timeline Standards:** OpenColorIO (OCIO), ACES colour pipeline, OpenTimelineIO (OTIO) sequence interchange.

Conform & I/O

🎞 Editorial Turnover, Conform & I/O

- **Conform & Turnover:** Package inspection, automated DI discrepancy reporting, regex pattern generation, multi-track Hiero conform.
- **Grade & Asset Processing:** CDL grades, shot LUT processing, frame range validation, USD plate sublayering.

Core Systems

Core Software & Systems Architecture

- **Languages & UI:** Python 3.x (Expert), PyQt / PySide (UI/UX engineering), C++, Java, Objective-C, Cocoa, shell scripting.
- **Infrastructure:** Docker, Linux/Unix environments, REST APIs, FastAPI, Git, Tractor render farm management & dispatching.

Automation

Production Automation & Integration

- **n8n Workflow Hub:** Inbound webhooks, reverse webhooks, asynchronous event routing, production API synchronisation.
- **Event Automation:** Automated show onboarding, smart plate validation, automated ShotGrid metadata injection.

💼 Professional Experience

FOUNDER • AI EDTECH PRODUCT

Aug 2026 - Present

Founder, Learn with Charlie

Charlie AI • London, UK

Focus: Multi-Agent Coding Teams, SymPy Verified Answers, Voice AI & React/FastAPI Stack

- **AI Agent Engineering & Governance:** Directed a team of specialised AI coding agents in Claude Code (builders, test reviewer, maths reviewer, and 3 simulated student agents testing in Playwright) under 50 ADRs, versioned data contracts, CI, and owner-approved merges.
- **Safety-First Verified Maths Model:** Engineered a zero-guessing architecture where the AI never decides an answer; every worked step is verified by a symbolic SymPy engine over parameterised templates, serving only human-approved teaching content.
- **Conversational Voice & Live Blackboard:** Designed natural voice-led instruction with ElevenLabs and a synchronised blackboard rendering steps in KaTeX and SVG, featuring button-led interaction ("I don't get it", "Make it simpler") with zero open chatbox risks.
- **Adaptive Gap Detection & Marking:** Architected two-question diagnostic probes detecting earlier-class foundational gaps (signs, fractions) with targeted micro warm-ups, line-by-line exercise marking that pinpoints errors, and end-of-topic checks.
- **Full-Stack Architecture & Brand:** Built with React, TypeScript, FastAPI (Python 3.12), PostgreSQL, Docker and Caddy, with country-based pricing on Cloudflare Workers, privacy by design, and licensing outreach to Indian textbook publishers.

CURRENT LEADERSHIP ROLE

Sep 2025 - Present

Lead Software Architect

Astra Studios • London, UK

Focus: Generative AI, Creative Systems & Production Technology

- **Engineering & Technical Leadership:** Lead Development and TD teams, defining production technology architecture and engineering standards while working directly with artists, production, supervisors and leadership to translate creative requirements into scalable technical systems.
- **Studio.AI GenAI Platform & Video Workflows:** Engineered the Studio.AI generative platform from architecture to production, integrating prompt-to-video engines (**Google Veo, Kling AI, Higgsfield, Seedance and ComfyUI node graphs**) with character identity consistency, artist-facing controls, and ACES colour conforming.
- **Production VFX & OpenUSD Pipeline:** Architected an end-to-end production-grade OpenUSD pipeline bridging DCCs (Maya, Nuke) from modelling to compositing for full-CG and live-action VFX.
- **Editorial Turnover & Conform I/O:** Engineered the studio's Editorial Turnover, Conform & I/O pipeline from scratch, integrating turnover package inspection, automated DI discrepancy reporting, regex pattern generation, multi-track Hiero conform, CDL grades, shot LUTs, and USD plate sublayering.
- **Zero-Touch Studio Automation:** Orchestrated zero-touch n8n studio automation hubs using inbound and reverse webhooks for automated show onboarding, smart plate validation, and ShotGrid metadata injection.

R&D Supervisor

Mar 2023 - Feb 2025

Technicolor Group • London, UK

- **Global Review Platform Architecture:** Spearheaded the architecture and productionisation of ReviewTool, a multi-facility VFX review platform with real-time 4K+ media caching, synchronisation and RV integration supporting distributed production workflows across global studios.
- **ProNuke Application Ecosystem:** Architected and engineered a modular Nuke production ecosystem spanning automated shot assembly, sequence-level publishing and batch render orchestration across global artist teams.
- **Meshroom Photogrammetry in Production:** Productionised Meshroom into matchmove and tracking workflows, cutting camera extraction turnaround from days to hours with fast switching between Meshroom, Maya, and 3D Equalizer.
- **Global R&D Leadership:** Led globally distributed R&D software engineering teams across London, Montreal, and Bengaluru supporting Compositing, Roto, Paint Prep, Camera Tracking, and Rotoanim departments.
- **Conform Ingest Automation:** Supervised Conform Ingest workflows using Hiero and Nuke Studio, automated CDL and LUT grade processing, and ShotGrid frame range validation.

R&D Supervisor

Jan 2022 - Feb 2023

Technicolor Group • Bengaluru, India

- **2D Software Engineering:** Headed 2D software engineering and pipeline development across Bengaluru and Montreal hubs.
- **Review Platform Prototyping:** Architected core compositing automation tools and early prototypes of the studio VFX review platform (ReviewTool).
- **Agile Execution:** Mentored engineers, established Agile and Scrum sprint frameworks, and aligned dev priorities with production executives.

Lead Pipeline Software Developer

Jul 2017 - Dec 2021

MPC Film • Bengaluru, India

- **Feature Production Pipelines:** Led core 2D pipeline solutions across Academy Award-winning films and global tentpole productions (1917, RRR, The Lion King).
- **Cross-Studio Platform Standardisation:** Spearheaded RPM Unification across MPC, MR.X and The Mill, consolidating and standardising software toolsets and UI/UX across compositing and tracking workflows to improve technical consistency across multiple studio environments.
- **Nuke Bridge Hub:** Developed Nuke Bridge integrating Tessa (AMS), ShotGrid (PMS), and Tractor (Render Farm), establishing a version-controlled, high-throughput pipeline.
- **Render Dispatcher:** Engineered the Nuke Render Dispatcher with automated render order traversal and Reserve & Commit dispatch mechanics.

Team Lead – Technology & R&D Production

Jul 2012 - Jun 2017

Technicolor • Bengaluru, India

- **Media Engines:** Architected macOS AVFoundation and QuickTime media processing engines (Channel Mapper, Frame Rate Converter, Compliance Edit Lists).
- **Team Management:** Managed an engineering team of 7 developers across C++, Objective-C, Cocoa, and Java, optimising production compression and authoring pipelines.
- **Hardware Compliance:** Served as primary technical contact for player manufacturers, resolving Blu-ray authoring framework and playback compatibility issues.

Senior Software Engineer – Technology

May 2008 - Jun 2012

Technicolor • Bengaluru, India • Los Angeles, USA • Hannover, Germany

- **Disney Collaboration (Los Angeles):** Collaborated on-site with Disney engineering in Los Angeles to architect the Fast Menu Load (FML) template and integrate Generation 2 Bootloaders (G2B).
- **World First 3D Blu-ray (Hannover):** Engineered core 3D capabilities into the BD-J Authoring framework at Technicolor R&I in Hannover, Germany, as a key contributor to the world's first released 3D Blu-ray disc.
- **Founding Engineer:** Founding software engineer at Technicolor India, building runtime alignment engines ('Graphics Adjuster') and studio authoring templates.

Application Engineer & Game Programmer

Sep 2006 - Apr 2008

e4e & Dhruva Interactive • Bengaluru, India

- Ported mobile game titles to Nokia, Motorola, Samsung, and Sony Ericsson devices for T-Mobile and Verizon; engineered touchscreen interfaces for early mobile OS platforms.

Honours, Awards & Fellowship

Technicolor Fellowship Network

2015 - 2025

Associate Member • Technicolor Group

> *"Awarded for production environment expertise, strong customer relationships, and deep collaboration with Research & Innovation—delivering a rare mix of creativity, pragmatism, and architectural stewardship."*
>
> — Fellowship Nomination Committee • Mar 2015

Fellowship Committee Election R&I Collaboration

Architectural Leadership

Studio Leadership & Excellence

2008 - 2024

9+ Honours • Technicolor, MPC Film & e4e

- **Value Champions & Best Employee of the Month (Aug 2019)** — *Visionary, Focused, Quietly Brilliant (Technicolor FEV)*
- **Performer of the Quarter • Team Lead (Oct 2016)** — *Technology R&D leadership*
- **Performer of the Quarter (2011, 2012, 2014)** — *Disney collaboration & pipeline innovation*
- **Extra Miler & Executive Team Awards (2011, 2013)** — *DVD/Blu-ray software & dedication*

Recent AI Professional Development

AI Generalist Accelerator Program

Generative AI Business Applications • Outskill

View Certificate ↗

> *14-Day Intensive Generative AI Accelerator mastering practical automation, MCP server tool deployment, AI agent systems, and no-code product development.*
>
> — Outskill • Aug 2026

Prompt Engineering n8n Workflow Automation

MCP Tool Integration Voice Agent Orchestration

Multimodal AI & Cloning

AI Accelerator C13 Hackathon

View Award ↗

2-Day Innovation Challenge • Outskill

> *Awarded for rapid prototyping, creativity, and deployment of functional generative AI application architectures during the 2-Day AI Innovation Challenge.*
>
> — Outskill Hackathon Jury • Aug 2026

Rapid AI Prototyping Agent System Architecture

Live Product Deployment

Education & Academic Background

Bachelor of Technology (B.Tech) • Computer Science & Engineering

2002 - 2006

Jawaharlal Nehru Technological University (JNTU) • Sri Venkateswara College of Engineering & Technology (SVCET)